# Table 1. Model comparison, posterior predictive behaviour and burden prediction

*All fits conditional on the calibrated mirror component. Burden columns are calibrated on the observed deletion coefficient, so the duplication prediction is out of sample.*

| Trait | Model | k | q | $w_{DEL}$ | $shift_{DEL}$ | $shift_{DUP}$ | ΔAIC | ΔBIC | ΔLOO | PPC DUP mean | PPC corr. | genes per Mb | pred. DUP burden | obs. DUP burden | pred/ obs | implied affected |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Height | Mirror alone | 0 | 0.0000 | | 0.0000 | 0.0000 | 77.23 | 72.74 | 38.75 | 0.000 | 0.370 | 4.61 | +0.1440 | -0.0784 | -1.84 | |
| Height | Continuous diagonal | 3 | 0.0817 | | 0.0049 | 0.0556 | 7.71 | 7.71 | 2.73 | 0.087 | 0.013 | 4.45 | -0.0223 | -0.0784 | 0.28 | |
| Height | Sparse hit diagonal | 3 | 0.0195 | | 0.0246 | 0.2816 | 5.26 | 5.26 | 1.85 | 0.052 | 0.007 | 4.45 | -0.0182 | -0.0784 | 0.23 | 0.364 |
| Height | Continuous axis-aligned | 4 | 0.0811 | 0.000 | 0.0000 | 0.0578 | 8.65 | 10.15 | 2.20 | 0.126 | 0.011 | 4.61 | -0.0286 | -0.0784 | 0.37 | |
| Height | Sparse hit axis-aligned | 4 | 0.0682 | 0.486 | 0.0705 | 0.2884 | 4.62 | 6.11 | 1.52 | 0.083 | 0.037 | 4.14 | -0.0514 | -0.0784 | 0.66 | 0.727 |
| Height | Continuous buffering | 3 | 0.0764 | | -0.0420 | 0.0734 | 2.57 | 2.57 | 0.76 | 0.341 | 0.054 | 6.47 | -0.0875 | -0.0784 | 1.12 | |
| Height | Sparse hit buffering | 3 | 0.0191 | | -0.1954 | 0.3483 | 0.00 | 0.00 | 0.00 | 0.191 | 0.096 | 6.32 | -0.0744 | -0.0784 | 0.95 | |
| Height | Buffering, saturating occ. | 4 | 0.0270 | | -0.1916 | 0.3456 | 1.75 | 3.25 | 0.54 | 0.213 | 0.094 | 6.54 | -0.1054 | -0.0784 | 1.34 | |
| Height | Aggregate drift (2 par) | 2 | | | 0.0000 | 0.0036 | 35.91 | 34.42 | 21.66 | 0.009 | 0.096 | 4.61 | +0.0115 | -0.0784 | -0.15 | |
| BMI | Mirror alone | 0 | 0.0000 | | 0.0000 | 0.0000 | 4.04 | 1.11 | 2.13 | 0.846 | 0.152 | 9.31 | -0.0701 | +0.0315 | -2.22 | |
| BMI | Continuous diagonal | 3 | 0.8000 | | 0.0009 | 0.0025 | 2.08 | 3.54 | 0.00 | 0.333 | 0.102 | 7.61 | +0.0137 | +0.0315 | 0.44 | |
| BMI | Sparse hit diagonal | 3 | 0.0426 | | 0.0254 | 0.0741 | 3.81 | 5.28 | 1.04 | 0.402 | 0.115 | 7.62 | +0.0168 | +0.0315 | 0.53 | 0.589 |
| BMI | Continuous axis-aligned | 4 | 0.8000 | 0.218 | 0.0039 | 0.0032 | 4.09 | 7.03 | 0.12 | 0.341 | 0.099 | 7.69 | +0.0145 | +0.0315 | 0.46 | |
| BMI | Sparse hit axis-aligned | 4 | 0.0422 | 0.111 | 0.1335 | 0.0881 | 5.88 | 8.81 | 1.16 | 0.368 | 0.113 | 8.25 | +0.0224 | +0.0315 | 0.71 | 0.586 |
| BMI | Continuous buffering | 3 | 0.8000 | | 0.0009 | 0.0025 | 2.08 | 3.54 | 0.22 | 0.356 | 0.108 | 7.61 | +0.0137 | +0.0315 | 0.44 | |
| BMI | Sparse hit buffering | 3 | 0.0426 | | 0.0254 | 0.0741 | 3.81 | 5.28 | 1.15 | 0.452 | 0.107 | 7.62 | +0.0168 | +0.0315 | 0.53 | |
| BMI | Buffering, saturating occ. | 4 | 0.0426 | | 0.0254 | 0.0741 | 5.81 | 8.74 | 1.15 | 0.408 | 0.113 | 7.62 | +0.0168 | +0.0315 | 0.53 | |
| BMI | Aggregate drift (2 par) | 2 | | | 0.0008 | 0.0020 | 0.00 | 0.00 | 0.12 | 0.349 | 0.107 | 7.50 | +0.0134 | +0.0315 | 0.43 | |

k, added parameters. q, per-gene event frequency (not identified for the continuous BMI models). $w_{DEL}$, fraction of events on the deletion axis. Shifts in trait SD. Deltas are relative to the best model within each trait. PPC values are posterior predictive CDFs; below 0.025 or above 0.975 is a failure. Burden in cm/Mb (height) and $kg/m^2$/Mb (BMI); observed values from Mace et al. 2017 Supplementary Table 1. Implied affected fraction applies to the sparse models only.